



illuminati - 2019

Advanced Mathematics Assignment-2B

CONIC SECTION

Section 1	Single Correct Answer Type
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Each of the following Question has 4 choices A, B, C & D, out of which ONLY ONE Choice is Correct.

- If focus of a parabola is $(1, 2)$ and feet of perpendiculars on any two tangents to this parabola from focus are $(3, 4)$ and $(4, 6)$ then vertex of the parabola is :
 (A) $\left(\frac{8}{5}, \frac{9}{5}\right)$ (B) $\left(\frac{7}{5}, \frac{9}{5}\right)$ ~~(C) $\left(\frac{9}{5}, \frac{8}{5}\right)$~~ (D) $\left(\frac{9}{5}, \frac{7}{5}\right)$
- The co-ordinates $(2, 3)$ and $(1, 5)$ are the foci of an ellipse which passes through the origin, then the equations of tangent and normal at origin are respectively :
 (A) $(3\sqrt{2} - 5)x + (1 - 2\sqrt{2})y = 0$ and $(3\sqrt{2} + 5)x - (2\sqrt{2} + 1)y = 0$
 (B) $(3\sqrt{2} + 5)x + (1 + 2\sqrt{2})y = 0$ and $(1 + 2\sqrt{2})y - (3\sqrt{2} + 1)y = 0$
 (C) $(1 - 2\sqrt{2})x + (3\sqrt{2} - 5)y = 0$ and $(3\sqrt{2} - 5)x - (1 - 2\sqrt{2})y = 0$
 (D) None of these.
- The length of latus rectum of the parabola passing through the points $(4, 4)$, $(6, 8)$, $(2, 4)$ and $(0, 8)$ is :
~~(A) 2~~ (B) $\frac{1}{2}$ (C) 7 (D) None of these
- A chord is drawn from a point $P(1, t)$ to the parabola $y^2 = 4x$ which cuts the parabola at A and B . If $PA \cdot PB = 3|t|$, then the range of t is :
 (A) $[2, 4]$ (B) $[1, 4]$ (C) $[-4, -1] \cup [1, 4]$ (D) None of these
- From any point P on the curve $b^4x + 2a^2y^2 = 0$, pair of tangents PQ and PR are drawn on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$. If QR touches a fixed parabola then the equation of the parabola is :
 (A) $y^2 = 4x$ (B) $y^2 = 8x$ (C) $x^2 = 4y$ (D) $x^2 = 8y$
- The locus of the focus of the family of parabolas having directrix of slope m and touching the lines $x = a$ and $y = b$ is :
 (A) $y + mx = b$ (B) $y + mx = a$ (C) $y + mx = bm + a$ (D) None of these
- If a variable normal of positive slope of the circle $x^2 + y^2 - 4x - 16y + 48 = 0$ intersects the parabola $y^2 = 4x$ at A and B , then the range of values of length of AB is :
 (A) $[0, 4]$ (B) $[0, 3\sqrt{2}]$ (C) $[0, 4\sqrt{2}]$ (D) None of these

8. An ellipse has the points $(1, -1)$ and $(2, -1)$ as its foci and $x + y - 5 = 0$ as one of its tangent. The coordinates of the point where this line touches the ellipse :
- (A) $\left(\frac{34}{9}, \frac{11}{9}\right)$ (B) $\left(\frac{34}{7}, \frac{1}{7}\right)$ (C) $\left(\frac{12}{5}, \frac{13}{5}\right)$ (D) None of these
9. If the normals to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ at the extremities of the chords $lx + my = 1$ and $px + qy = 1$ are concurrent then :
- (A) $a^2lp = b^2mq$ (B) $a^2lp + b^2mq = 0$
 (C) $a^2lp = 2b^2mq$ (D) None of these
10. If m is slope of all the tangents to the parabola $y^2 = 4x$ which bisect two distinct chords of the ellipse $\frac{x^2}{4} + \frac{y^2}{1} = 1$ drawn from $(-2, 0)$ then :
- (A) $(-\infty, -1) \cup (1, \infty)$ (B) $\left(-\infty, -\frac{2}{3}\right) \cup \left(\frac{2}{3}, \infty\right)$
 (C) $(-1, 1)$ (D) $\left(-\frac{2}{3}, \frac{2}{3}\right)$
11. An ellipse slides between two given lines which are at right angles to each other. The locus of the centre of the ellipse is :
- (A) Ellipse (B) Parabola (C) Circle (D) Pair of straight line
12. The directrices of the ellipse $3(x + y - 5)^2 + 2(x - y + 7)^2 = 12$ is :
- (A) $x - y = 1$ (B) $x - y = 12$ (C) $x - y + 13 = 0$ (D) None of these
13. If the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is inscribed in a rectangle whose length to breadth ratio is $2 : 1$ then the area of rectangle is :
- (A) $\frac{a^2 + b^2}{5}$ (B) $\frac{2}{5}(a^2 + b^2)$ (C) $\frac{8}{5}(a^2 + b^2)$ (D) $\frac{a^2 + b^2}{3}$
14. If I_1 and I_2 be the length of the focal chord other than latus rectum of ellipse $\frac{x^2}{4} + \frac{y^2}{3} = 1$ which are perpendicular to each other then $\frac{1}{I_1} + \frac{1}{I_2} =$
- (A) $\frac{12}{7}$ (B) $\frac{7}{12}$ (C) $\frac{5}{12}$ (D) None of these
15. The range of eccentricity of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ ($a > b$) such that the line segment joining the foci does not subtend a right angle at any point on the ellipse :
- (A) $\left(0, \frac{1}{\sqrt{3}}\right)$ (B) $\left(0, \frac{1}{\sqrt{2}}\right)$ (C) $\left(0, \frac{1}{3}\right)$ (D) $\left(0, \frac{1}{2}\right)$
16. A circle whose centre lies on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ intersects the rectangular hyperbola $xy = c^2$ in four points. Then the locus of the centre of the locus of the centroid of triangle formed by any three points of intersection is :
- (A) Ellipse (B) Parabola (C) Circle (D) Hyperbola

17. The locus of focus of family of hyperbolas having fixed centre and two fixed perpendicular lines as its tangents is :
 (A) Ellipse (B) Parabola
 (C) Pair of straight line (D) Rectangular Hyperbola
18. Two concentric rectangular hyperbolas are such that the foci of one lies on the other. Then the angle between their axes is :
 (A) $\frac{\pi}{3}$ (B) $\frac{\pi}{2}$ (C) $\frac{\pi}{6}$ (D) $\frac{\pi}{4}$
19. Two tangents to a hyperbola meet its axes in concyclic points. Then the locus of their point of intersection is :
 (A) Ellipse (B) Parabola
 (C) Pair of straight line (D) Rectangular Hyperbola
20. The length of the latus rectum for the rectangular hyperbola $xy = \pi$ is :
 (A) $2\sqrt{2}\pi$ (B) $\sqrt{2}\pi$ (C) $2\sqrt{2}\pi$ (D) None of these
21. The normals at points P, Q, R on a rectangular hyperbola intersect at a point S on the curve. Then the centre of hyperbola isof the triangle PQR .
 (A) Centroid (B) Ortho center (C) Circum center (D) In center
22. A tangent is drawn at a point (x_1, y_1) on the parabola $y^2 = 4ax$. Now tangents are drawn from points on this tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ such that all the chords of contact pass through the point (x_2, y_2) then :
 (A) $x_1 x_2 > 0$ (B) $x_1 x_2 < 0$ (C) $x_1 x_2 = 2a$ (D) None of these
23. The number of points on rectangular hyperbola $y^2 - x^2 = a^2$, such that their chord of contact with respect to $xy = c^2$ is normal to curve $xy = c^2$ is :
 (A) 0 (B) 1 (C) 2 (D) 4
24. If the equation the family of the ellipse is $\frac{x^2}{\cos^2 \alpha} + \frac{y^2}{\sin^2 \alpha} = 1 \left(0 < \alpha < \frac{\pi}{4} \right)$ then the locus of the extremities of the latus rectum is :
 (A) $2y(1 - x^2) = 1 + x^2$ (B) $2y^2(1 + x^2) = (1 - x^2)^2$
 (C) $y^2 = (1 - x^2)^2$ (D) None of these
25. From a point on $4x + 3y = 25$ two mutually perpendicular tangents are drawn to $\frac{x^2}{16} + \frac{y^2}{9} = 1$ and from their points of contact normals are drawn which intersect at point P . The coordinates of P is :
 (A) $(0, 0)$ (B) $\left(2, \frac{3}{2}\right)$ (C) $\left(-2, -\frac{3}{2}\right)$ (D) None of these
26. Let P be any point on any directrix of an ellipse. Then chords of contact of point P w.r.t. the ellipse and its auxiliary circle intersect at :
 (A) some point on the major axis depending upon the position of point P
 (B) mid point of the line segment joining the centre to the corresponding focus
 (C) corresponding focus
 (D) None of these

27. Let P be any point on a directrix of an ellipse of eccentricity e . S be the corresponding focus and C the centre of the ellipse. The line PC meets the ellipse at A . The angle between PS and tangent at A is α then α , is equal to :
- (A) $\tan^{-1} e$ (B) $\frac{\pi}{2}$ (C) $\tan^{-1}(1-e^2)$ (D) None of these
28. If tangents PQ and PR are drawn from a point on the circle $x^2 + y^2 = 25$ to the ellipse $\frac{x^2}{16} + \frac{y^2}{b^2} = 1$, ($b < 4$), so that the fourth vertex S of parallelogram $PQSR$ lies on the circumcircle of triangle PQR , then eccentricity of the ellipse is :
- (A) $\frac{\sqrt{5}}{4}$ (B) $\frac{\sqrt{7}}{3}$ (C) $\frac{\sqrt{7}}{4}$ (D) $\frac{\sqrt{5}}{3}$
29. If the ellipse $\frac{x^2}{a^2-7} + \frac{y^2}{13-5a} = 1$ is inscribed in a square of side length $\sqrt{2}$ then a is equal to :
- (A) $\frac{6}{5}$ (B) $(-\infty, -\sqrt{7}) \cup (\sqrt{7}, 13/5)$
 (C) $(-\infty, -\sqrt{7}) \cup (13/5, \sqrt{7})$ (D) no such 'a' exists
30. If foci of hyperbola lie on $y = x$ and one of the asymptote is $y = 2x$, then equation of the hyperbola, given that it passes through $(3, 4)$, is :
- (A) $x^2 - y^2 - \frac{5}{2}xy + 5 = 0$ (B) $2x^2 - 2y^2 + 5xy + 5 = 0$
 (C) $2x^2 + 2y^2 - 5xy + 10 = 0$ (D) None of these
31. The locus of the centre of a variable circle touching two circles of radius r_1 and r_2 externally, which also touch each other externally, is a conic. The eccentricity of the conic, if $\frac{r_1}{r_2} = 3 + 2\sqrt{2}$, is :
- (A) 1 (B) $\sqrt{2}$ (C) $1/\sqrt{2}$ (D) $2\sqrt{2}$
32. Four points are such that the line joining any two points is perpendicular to the line joining other two points. If three points out of these lie on a rectangular hyperbola then the fourth point will lie on :
- (A) the same hyperbola (B) conjugate hyperbola
 (C) one of the directrix (D) one of the asymptotes
33. Let P be the point of intersection of the hyperbolas $xy = c^2$ and $x^2 - y^2 = a^2$ in the first quadrant. If tangents at P to both the curves intersect the y -axis at Q and R then circumcentre of triangle PQR lies on :
- (A) $x + y = 1$ (B) $x - y = 1$ (C) x -axis (D) y -axis
34. The circle drawn with variable chord $x + ay - 5 = 0$ (a being a parameter) of the parabola $y^2 = 20x$ as diameter will always touch the line
- (A) $x + 5 = 0$ (B) $y + 5 = 0$ (C) $x + y + 5 = 0$ (D) $x - y + 5 = 0$
35. Two parabolas have the same focus $(3, -2)$. Their directrices are the x -axis and the y -axis respectively. Then the slope of their common chord is :
- (A) -1 (B) $-\frac{1}{2}$ (C) $-\frac{\sqrt{3}}{2}$ (D) None of these

36. The locus of the point which moves such that its distance from the point $(4, 5)$ is equal to its distance from the line $7x - 3y - 13 = 0$ is :
 (A) a straight line (B) a circle (C) a parabola (D) None of these
37. If bisector of the angle APB where PA and PB are the tangents to the parabola $y^2 = 4ax$ is equally inclined to the coordinate axes, then the point P lies on :
 (A) tangent at vertex of the parabola (B) directrix of the parabola
 (C) circle with centre at origin and radius a (D) the line of latus rectum
38. A variable parabola of fixed latus rectum $4b$ and having axis parallel to x -axis, lies completely in Ist and IVth quadrant and cuts the fixed parabola $y^2 = 4ax$ orthogonally. The locus of vertex of the variable parabola is :
 (A) $y^2 = 4ab$ (B) $y^2 = 16ab$ (C) $x^2 = 4ab$ (D) None of these
39. If two distinct tangents can be drawn from the point $(\alpha, 2)$ on different branches of the hyperbola $\frac{x^2}{9} - \frac{y^2}{16} = 1$, then the range of α is :
 (A) $\left(\frac{-1}{2}, \frac{1}{2}\right)$ (B) $\left(\frac{-3}{2}, \frac{3}{2}\right)$ (C) $(-1, 1)$ (D) None of these

Section 2

Link Comprehension Type

Each of the following Question has 4 choices A, B, C & D, out of which ONLY ONE Choice is Correct.

Paragraph for Questions 40 - 43

A conic section is obtained by the intersection of two inverted cones (having same vertex) with a plane. If a plane passes through the vertex, then its intersection with the cones either represents a pair of straight lines or a point depending upon whether it intersects the cones or not. If plane does not pass through vertex, then section may be hyperbola or circle depending upon whether plane is parallel to axis of the cone or perpendicular to it. If it has any other inclination, then intersection of plane and cone gives an ellipse. If a variable point P moves such that the line passing through P and $Q(0, 0, 2)$ makes an angle 60° with z -axis, then

40. The locus of P is :
 (A) $x^2 - y^2 - 3(z - 2)^2 = 0$ (B) $x^2 - y^2 + 3(z - 2)^2 = 0$
 (C) $x^2 + y^2 - 3(z - 2)^2 = 0$ (D) $x^2 + y^2 + 3(z - 2)^2 = 0$
41. The locus of intersection of locus of P and the plane $x + y + z = 2$ is :
 (A) a pair of straight lines (B) a hyperbola
 (C) a circle (D) a point
42. The locus of intersection of locus of P with $x + y = 2$ is :
 (A) a straight line (B) a hyperbola (C) a circle (D) an ellipse
43. The locus of intersection of locus of P and the plane $x + y + z = 1$ is :
 (A) a pair of straight lines (B) a hyperbola
 (C) a circle (D) an ellipse

Paragraph for Questions 44 - 48

If $ax^2 + by^2 + 2hxy + 2gx + 2fy + c = 0$ represents an ellipse, then $h^2 < ab$ and $abc + 2fgh - af^2 - bg^2 - ch^2 \neq 0$. If for every point (x_1, y_1) satisfying above equation $(2h - x_1, 2k - y_1)$ also satisfy it, then (h, k) is centre of it. The length of semi major axis and minor axis is nothing but the maximum and minimum value of the distance of a point lying on the curve from its centre.

For the ellipse $2x^2 - 2xy + 4y^2 - (3 + \sqrt{2}) = 0$ answer the following questions (1-5).

44. The inclination of major axis of it with x -axis is :

- (A) $\frac{\pi}{12}$ (B) $\frac{\pi}{8}$ (C) $\frac{3\pi}{8}$ (D) $\frac{5\pi}{8}$

45. The equation of tangent to ellipse such that sum of perpendiculars dropped from foci is 2 units, is :

- (A) $y \cos \frac{3\pi}{4} - x \sin \frac{3\pi}{4} = 1$ (B) $y \sin \frac{3\pi}{8} - x \cos \frac{3\pi}{8} = 1$
 (C) $x \cos \frac{\pi}{8} - y \sin \frac{\pi}{8} = 1$ (D) $y \cos \frac{5\pi}{8} + x \sin \frac{5\pi}{8} = 1$

46. The product of perpendicular from the foci to any tangent to above given ellipse is :

- (A) 4 (B) 2 (C) 1 (D) $\frac{1}{2}$

47. If the given ellipse is rotated so that it has its major axis coincident with x -axis, then its equation becomes :

- (A) $3(x^2 + y^2 - 1) = (y^2 - x^2 + 1)\sqrt{2}$ (B) $3(x^2 + y^2 - 1) = (y^2 - x^2 - 1)\sqrt{2}$
 (C) $3(x^2 + y^2 - 1) = (y^2 - x^2 - 1)\sqrt{2}$ (D) $\sqrt{2}(x^2 - y^2 + 1) = 3(x^2 + y^2 - 1)$

48. The maximum area of the circle lying inside the given ellipse is :

- (A) π (B) $\frac{\pi}{2}$ (C) $\pi \left(\frac{3 + \sqrt{2}}{3 - \sqrt{2}} \right)$ (D) $\pi \left(\frac{3 - \sqrt{2}}{3 + \sqrt{2}} \right)$

Section 3**Multiple Correct Answer Type**

Each of the following Question has 4 choices A, B, C & D, out of which ONE OR MORE Choices may be Correct.

49. For the conic $9x^2 + 16y^2 + 24xy - 104x + 28y - 124 = 0$.

- (A) Vertex at $\left(\frac{-16}{25}, \frac{37}{25} \right)$ (B) Length of latus rectum is 2 units.
 (C) Tangent at vertex is line $4x - 3y + 7 = 0$ (D) Equation of axis is $3x + 4y = 4$

50. For the conic $8x^2 + 5y^2 + 4xy + 16x + 4y + 3 = 0$.

- (A) Equation of major axis is $2x + y + 2 = 0$
 (B) Equation of minor axis is $x - 2y + 1 = 0$
 (C) eccentricity is $\frac{\sqrt{5}}{4}$ (D) length of latus rectum is $\frac{7}{9}$.

51. For the conic $7x^2 - 2y^2 + 12xy - 2x + 14y - 22 = 0$.

- (A) directrix is $2x + y = 1$ (B) focus is at $(1, 2)$
 (C) eccentricity is $\sqrt{2}$
 (D) equation of transverse axis is $x - 2y + 3 = 0$

52. $px^2 + 2qxy + ry^2 + 2sx + 2ty + u = 0$ (where $p \neq q \neq r$ and p, q, r, s, t and u are constants) will represent :

- (A) ellipse, if p, q, r are in H.P. (B) parabola, if p, q, r are in G.P.
 (C) hyperbola, if p, q, r are in A.P. (D) above are true only if $\begin{vmatrix} p & q & s \\ q & r & t \\ s & t & u \end{vmatrix} \neq 0$

53. If $\frac{3}{2 + e^{i\theta}} = \lambda x + i\mu y$, then locus of $P(x, y)$ will represent :

- (A) Ellipse if $\lambda = 1, \mu = 2$ (B) Pair of straight lines if $\mu = 0, \lambda = 1$
 (C) Circle if $\lambda = \mu = 1$ (D) Hyperbola if $\mu = 1, \lambda = -1$

54. A parabola is drawn with focus is at one of the foci of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, (a > b)$ and directrix passing through the other focus and perpendicular to the major axes of the ellipse. If latus rectum of the ellipse and the parabola are same, and 'e' be the eccentricity of ellipse then :

- (A) $e = \sqrt{2} - 1$ (B) $e = \sqrt{3} - 1$
 (C) length of latus rectum $= 2ae$ (D) length of latus rectum $= 4ae$

55. An ellipse with major and minor axes lengths as $2a$ and $2b$ touches coordinate axes in first quadrant and having foci (x_1, y_1) and (x_2, y_2) . If r is the radius of director circle of ellipse then :

- (A) $x_1 \cdot x_2 = b^2$ (B) $y_1 \cdot y_2 = a^2$
 (C) $r = \frac{1}{2} \sqrt{(x_1 + x_2)^2 + (y_1 + y_2)^2}$ (D) $r = \frac{1}{2} \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$

56. The locus of the image of the focus of the ellipse $\frac{x^2}{25} + \frac{y^2}{9} = 1$ with respect to any of the tangents to the given ellipse is a curve S . For the curve S , correct option(s) is/are :

- (A) S is circle with centre $(4, 0)$ & radius 10.
 (B) S is circle with centre $(-4, 0)$ & radius 10.
 (C) S is ellipse with centre $(4, 0)$ & length of major axes is 10
 (D) S is ellipse with centre $(-4, 0)$ & length of minor axes is 10

57. For all real p , the line $2px + y\sqrt{1 - p^2} = 1$ touches a fixed ellipse E , whose axes are coordinate axes. Choose the correct option(s) for the ellipse E .

- (A) eccentricity is $\frac{\sqrt{3}}{2}$ (B) foci are at $\left(0, \pm \frac{\sqrt{3}}{2}\right)$
 (C) foci are at $\left(\pm \frac{\sqrt{3}}{4}, 0\right)$ (D) radius of director circle of E is $\frac{\sqrt{5}}{4}$

58. A coplanar beam of light emerging from a point source has the equation $\lambda x - y + 2(1 + \lambda) = 0$, $\lambda \in R$, the rays of the beam strike an elliptical surface and get reflected. The reflected rays from another convergent beam having equation $\mu x - y + 2(1 + \mu) = 0$, $\mu \in R$. Further it is found that the foot of perpendicular from the point $(2, 2)$ upon any tangent to the ellipse lies on the circle $x^2 + y^2 - 4y - 5 = 0$. If a , b and e are the semi major axis, semi minor axis and eccentricity of the ellipse respectively then :

~~(A)~~ $a = 3$ (B) $b = 5$ ~~(C)~~ $e = \frac{2}{3}$ ~~(D)~~ foci $(\pm 2, 2)$

59. The ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is such that it has the least area but contains the circle $(x - 1)^2 + y^2 = 1$ then for the ellipse :

(A) eccentricity is $\sqrt{\frac{2}{3}}$
 (B) Equation of auxiliary circle is $x^2 + y^2 = 4.5$
 (C) Length of latus rectum is 4
 (D) foci of the ellipse are at $(\pm\sqrt{3}, 0)$

60. If a number of ellipse be described with same major axis $2a$ but variable minor axis then the tangents at the ends of their latus rectum pass through fixed points (α, β) and the extremities lies on the curve $x^2 = a(a + py)$ then :

(A) $\alpha = 0, \beta = \pm a$ (B) $\alpha = \pm a, \beta = 0$ (C) $p = \pm 1$ (D) $p = \pm 2$

61. If a pair of variable straight lines $x^2 + 4y^2 + \alpha xy = 0$ (where, α is real parameter) cuts the ellipse $x^2 + 4y^2 = 4$ at two points A and B then, locus of the point of intersection of tangents at A and B is/are :

~~(A)~~ $x - 2y = 0$ (B) $2x - y = 0$ ~~(C)~~ $x + 2y = 0$ (D) $2x + y = 0$

62. Let $y^2 = 4ax$ be a parabola and $x^2 - y^2 = a^2$ be a hyperbola. Then number of common tangents is :

~~(A)~~ 2 for $a < 0$ (B) 1 for $a < 0$ ~~(C)~~ 2 for $a > 0$ (D) 1 for $a > 0$

63. Let P and Q be the extremities of a chord of the parabola $y^2 = 4ax$. If the circle with PQ as its diameter touches the directrix of the parabola at T then :

(A) the abscissa of P , T and Q are in G.P. (B) the abscissa of P , T and Q are in A.P.
 (C) the ordinates of P , T and Q are in A.P. (D) the ordinates of P , T and Q are in G.P.

64. Let a parabola with focus as the vertex A of one segment of hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ with centre O and vertex as the focus S of that segment. If the parabola cuts positive y-axis at K then :

~~(A)~~ OK is geometric mean of OS and the latus rectum of parabola.
 (B) OK is harmonic mean of OS and the latus rectum of parabola.
~~(C)~~ Equation of parabola is $y^2 = -4a(e - 1)(x - ae)$
~~(D)~~ Vertex of parabola is $(ae, 0)$

Section 4

Single Integer Answer Type

Each of the following question has an integer answer between 0 and 9.

65. The locus of a point whose chord of contact with respect to the ellipse $x^2 + 2y^2 = 1$ subtends a right angle at the centre of the ellipse is $ax^2 + by^2 + c = 0$ then $a + b + c = \dots\dots\dots$ **2**
66. A normal is drawn to the ellipse $\frac{x^2}{a^2 + 2a + 2} + \frac{y^2}{a^2 + 1} = 1$ whose centre is at O . If maximum radius of the circle, centered at the origin and touching the normal, is 5 then non negative values of ' a ' is $\dots\dots\dots$
67. The lengths of two perpendicular focal chords of the parabola $y^2 = 4(x+1)$, are p and q . Then the value of $\frac{pq}{p+q}$ is $\dots\dots\dots$ **4**
68. Let PQ be the common chord of the parabola $y^2 = 4ax$ and the circle touching the parabola at P and Q . Tangent at P and the chord PQ are equally inclined to the axes of the parabola. The locus of middle point of PQ is another parabola whose latus rectum is $\frac{4a}{k}$ then $k = \dots\dots\dots$
69. The values of $[\alpha]$ for which three distinct chords drawn from $(\alpha, 0)$ to the ellipse $x^2 + 2y^2 = 1$ are bisected by the parabola $y^2 = 4x$ is : $([.] = \text{G.I.F}) \dots\dots\dots$
70. If AB and CD are two perpendicular focal chords of a parabola $y^2 = 28x$. If a quadrilateral $ABCD$ is formed then the minimum area of the quadrilateral is $2^a \cdot 7^b$ then $a + b = \dots\dots\dots$ **1**
71. The radius of largest circle with centre $(3, 0)$ that can be inscribed in the ellipse $9x^2 + 25y^2 = 225$ is $\frac{a\sqrt{b}}{c}$ (G.C.F. $(a, c) = 1$) then $a + b - c = \dots\dots\dots$ **6**
72. Two perpendicular tangents from a point on the circle $x^2 + y^2 = 9$ are drawn to an ellipse (eccentricity is e) If the chord of contact of these tangents touches the ellipse $x^2 + 4y^2 = 4$ Then $\frac{1}{e^2} = \dots\dots\dots$ **2**
73. If a chord of hyperbola $xy = c^2$ is a normal at point A , subtending an angle α at origin. Then $\frac{\sin(\alpha - A)}{\sin(\alpha + A)} = \dots\dots\dots$
74. At the point of intersection of rectangular hyperbola $xy = -c^2$ and parabola $x^2 = 4by$, tangents to the hyperbola and the parabola make angle θ_1 and θ_2 respectively with the x -axis. Then $2 \tan \theta_1 + \tan \theta_2 = \dots\dots\dots$ **0**

ANSWERS KEY

Advanced Mathematics Assignment-2B | Conic Section

Section 1	Single Correct Answers Type
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1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
C	A	A	C	B	D	C	A	A	B	C	C	C	B	B
16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
D	D	C	D	C	A	B	C	B	A	C	B	C	D	C
31	32	33	34	35	36	37	38	39						
B	A	D	A	A	A	D	B	B						

Section 2	Link Comprehension Type
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40	41	42	43	44	45	46	47	48
C	A	B	D	B	B	C	D	A

Section 3	Multiple Correct Answer Type
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49	50	51	52	53	54	55	56	57	58	59
ACD	AB	ACD	ABCD	ABC	AD	AC	AB	AB	ACD	ABD
60	61	62	63	64						
AC	AC	AC	AC	ACD						

Section 4	Single Integer Answer Type
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65	66	67	68	69	70	71	72	73	74
2	2	4	5	8	7	6	2	3	0